



A Comparative Analysis of Different ML Models for Wearable Devices Centric Human Activity Recognition

CSE498R

Submitted To

Riasat Khan(Rtk)

Assistant Professor

**Department of Electrical and Computer Engineering
North South University**

Submitted By

Name	ID
Namia Jalal Zara	1911522042
Jarin Tasnim	1912608042
Rashiqur Rahman Rifat	1911445642
Md. Baker	1911672642

Declaration

This is to declare that no part of this report or the project has been previously submitted elsewhere for the fulfillment of any other degree or program. Proper acknowledgement has been provided for any material that has been taken from previously published sources in the bibliography section of this report.

.....
Namia Jalal Zara

ECE Department
North South University, Bangladesh

.....
Jarin Tasnim

ECE Department
North South University, Bangladesh

.....
Rashiqur Rahman Rifat

ECE Department
North South University, Bangladesh

.....
Md. Baker

ECE Department
North South University, Bangladesh

Approval

The Project entitled “A Comparative Analysis of Different ML Models for Wearable Devices Centric Human Activity Recognition ” by Namia Jalal Zara (1911522042), Jarin Tasnim (1912608042), Rashiquir Rahman Rifat (1911445642), Md. Baker (1911672642) has been accepted as satisfactory and approved for partial fulfillment of the requirement of BS in CSE degree program.

Supervisor’s Signature

Riasat Khan
Department of Electrical and Computer Engineering
North South University
Dhaka, Bangladesh.

Department Chair’s Signature

Dr. Rajesh Palit
Department of Electrical and Computer Engineering
North South University
Dhaka, Bangladesh.

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Abstract

This paper presents a comprehensive study on applying machine learning models for human activity recognition. The increasing number of wearable devices and the growing interest in health monitoring and smart environments have fueled the demand for robust and accurate systems that automatically recognize human activities. This study explores and compares the performance of different machine-learning models in human activity recognition. We aim to identify the most effective ML model to classify and predict human activities accurately. We evaluate various models, including traditional classifiers such as Decision Tree, Random Forest, K-Nearest Neighbor, Logistic Regression, AdaBoost algorithm, and XGBoost algorithm. All models have been evaluated using the publicly available dataset "Human Activity Recognition with Smartphones", which captures daily activities: Lying, Standing, Sitting, Walking, Walking Upstairs, and Walking Downstairs. Among all the models we applied, XG Boost gave the highest accuracy of 99.52% based on precision (100%), recall (100%), and F1 score (100%). We performed hyperparameter tuning on the ML models to attain the best accuracy.

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Chapter 1: Project Overview

1. Project Overview:

1.1. Introduction

Human Activity Recognition (HAR) provides insights into human behavior, exemplifying the combination of computer science, engineering, and human-computer interaction. Through the use of sensors built into wearable technology, HAR is possible to record human movements in a variety of contexts, including security, smart environments, healthcare, fitness, and industrial safety. Rich data streams are transmitted by these sensors, bringing about a new knowledge of human behavior that goes beyond simple data aggregation and signals a paradigm change in how we interact with the outside world.

Personalized therapies and proactive monitoring are made possible in the healthcare industry by HAR, which helps researchers and practitioners understand the complexity of patient health. It offers a comprehensive viewpoint on physical activity in fitness training, assisting people in reaching their highest level of wellbeing. Furthermore, HAR plays a role in the creation of smart environments by improving their capacity to adapt and respond to human presence in homes, workplaces, and urban areas, which increases sustainability, efficiency, and comfort.

Furthermore, through the detection of unusual activity, HAR plays a critical part in supporting security measures by recognizing and mitigating potential risks. With its insights into everything from industrial safety to intruder detection, HAR helps create a safer society where technology is a watchful steward of human wellbeing. Fundamentally, HAR embodies the mutually beneficial association between humans and machines, laying the groundwork for a future in which technology enhances our talents and broadens our shared experiences.

1.2. Motivation & Problem Statement:

- 1.2.1. **Motivation:** A new era of Human Activity Recognition (HAR) has begun with the widespread use of wearable devices with sensors, offering insights into human activity in a variety of disciplines. It's critical to comprehend the efficacy of various machine learning models for HAR as wearable technology becomes more widely used. This kind of analysis not only helps us better understand how people move, but it also makes it easier to create activity detection algorithms that are more precise and effective. This study compares and assesses the effectiveness of various machine learning models in an effort to determine which strategies are best for wearable device-centric HAR applications.

1.2.2. Problem Statement: The problem statement centers on the requirement to identify the best machine learning models for Human Activity Recognition (HAR) by employing wearable device data. The selection of proper machine learning algorithms is still a key challenge, even with the increased interest in HAR systems. Accurate and reliable activity recognition is hampered by a number of factors, including processing efficiency, sensor data unreliability, and activity pattern complexity. In order to determine the effectiveness, scalability, and applicability of various machine learning models for wearable device-centric HAR, a thorough comparative analysis of these models is urgently needed. By shedding light on the performance traits and trade-offs connected to different machine learning techniques, this research seeks to close this knowledge gap in the field of HAR and pave the way for the creation of increasingly complex and dependable activity recognition systems for wearable technology.

1.3. Proposed Solution:

1.3.1. Major components of the project: Identifying a project's key components is the first step in any analysis.. For our project, the major components we are using are described below:

- **Dataset Preprocessing:** Preparing the dataset for machine learning models is the task of this component. It involves activities including processing missing data, eliminating duplicates, and applying label encoding to transform category variables into numerical ones. It also involves checking for null values. To further normalize the data, feature scaling methods such as MinMaxScaler are used. To make the evaluation of the model easier, the dataset is then divided into training and testing sets.
- **Machine Learning Algorithms:** It focuses on the selection and application of several machine learning algorithms for the detection of human activities. Decision Tree, Random Forest, Logistic Regression, AdaBoost, K-Nearest Neighbors (KNN), and XGBoost are the six models used in the project. Using the preprocessed dataset, each algorithm is used to create prediction models.
- **Hyperparameter Tuning:** A crucial first step in maximizing the effectiveness of machine learning models is hyperparameter tuning. In this component, a variety of hyperparameters for each algorithm are methodically searched through utilizing methods such as GridSearchCV and RandomizedSearchCV. Finding the ideal set of hyperparameters to maximize model performance is the aim.
- **Explainable AI (XAI):** Machine learning models are made more interpretable through the use of Explainable AI (XAI) techniques like Local Interpretable Model-agnostic Explanations (LIME). By highlighting significant characteristics and offering insights into the decision-making process, these methods assist users in comprehending how the models make their predictions.
- **Evaluation and Comparison:** This part entails assessing each machine learning model's performance using measures including F1 score, accuracy, precision, and recall.

Performance of the models both with default hyperparameters and after hyperparameter modification is compared. To display the classification results for various activity classes, confusion matrices are created.

1.4. Project Goals:

The main goal of this project is to create a reliable Human Activity Recognition (HAR) system by using wearable technology with sophisticated machine learning algorithms and sensors. In real-time, the system seeks to precisely recognize and categorize a range of human actions, including standing, walking, running, sitting, and certain workouts. Through the utilization of sensor data and machine learning models, the project aims to provide a flexible and adaptive solution that can handle a range of applications, such as industrial safety, smart environment control, fitness tracking, healthcare monitoring, and security enhancement. Delivering a HAR system that is dependable, easy to use, and offers tailored feedback along with actionable insights and a smooth integration into daily life is the ultimate aim, as it promotes better health outcomes, safety, and user satisfaction.

1.5. Summary:

The project's goals are outlined in this section, with a particular emphasis on Human Activity Recognition (HAR) employing cutting-edge machine learning algorithms and wearable sensor technologies. It draws attention to how HAR is multidisciplinary and covers fields including security, fitness, healthcare, and smart environments. The rising popularity of wearable devices serves as a driving force behind the project's motivation to assess and develop machine learning models for activity identification, as this section highlights. It describes the project's objective of creating a dependable HAR system with real-time activity identification in order to improve user enjoyment, safety, and health outcomes by being seamlessly integrated into everyday activities.

Chapter 2: Related Work & Theory

2. Related Work & Theory:

2.1. Introduction:

In HAR systems, machine learning models like Ada Boost, XGBoost, KNN, Random Forest, Decision Tree, Logistic Regression, and others are frequently used to assess human activity based on information gathered from wearable devices. Personalized health monitoring and fitness training, smart settings that adapt, increased security via intruder detection, and increased industrial safety by detecting risky behaviors are just a few benefits that come with HAR.

2.2. Related Work:

Some notable existing papers in the market given below:

Mahajan [1] and colleagues studied human activity recognition from videos using K-mean and ANN. They used a private dataset and they made it using different devices. Most of that video had noises so they preprocessed activity videos. They also used Mean Squared Error(MSE) to evaluate the performances. By using Artificial Neural Network they had an accuracy of 98.24%.

Gumaei [2] and colleagues studied human activity from body sensors by using deep learning models like RNN(GPU, SRU). They used the MHEALTH dataset which is a public dataset. Body motion and different sins of 10 subjects were used in this dataset. They used precision, and F1-score to evaluate the performances. They had an accuracy of 99.80% for deep SRUs-GRUs and 94.145 for deep SRUs.

J. Liu et al. [3] studied new methodologies for top-to-bottom-based human movement examination and demonstrated the viability of 3D human action analysis. This paper presented a large-scale human activity recognition dataset (NTU RGB+D dataset, SBU Interaction dataset, UT-Kinect dataset, Berkeley MHAD dataset) including 56,000 video tests and 4 million edges collected from 40 subjects. Sixty activity classes, including daily, shared, and wellness-related activities, are available. They had 100% accuracy for the Berkeley MHAD dataset in HBRNN methods.

Calatrava-Nicolas et al. [4] studied human activity recognition (HAR) using a light residual network for wearable sensor data. Their goal of using Light Residual Network (ResNet) is to perform tasks such as modifying ResNet18 and reducing the complexity of the model and parameters. They used the UCI-HAR dataset, which contains different activities, and they focused on six activities. They compared their work with previous work F1-Score, accuracy, and number of parameters, and they got the best accuracy, 97.6%, for their model.

S. Ghazal [5] and his team studied human activity recognition using 2D skeleton data and supervised machine learning. Two publicly available datasets were used for the performance evaluation. They evaluated the performance of various classifiers, including Feed-forward, KNNs, SVM, Discriminant analysis, and Naive Bayes. Naive Bayes achieved 90% accuracy in the first set of experiments, followed by SVM at 76%. Linear Discriminant Analysis (LDA) excelled in classifying falls, while Naive Bayes and KNN achieved 100% recall for standing. After data normalization, all classifiers showed increased accuracy, with KNN reaching 98%.

N. S. Pathan [6] and his colleagues studied machine learning-based human activity recognition during physical exercise using wavelet packet transform of PPG and Inertial Sensors data. They used publicly available PPG data collected from the physionet. The study used an 8-second time series window with 75% overlap for frame-by-frame analysis of physiological and inertial data, using Wavelet Packet Transform for feature extraction. The Random Forest classifier achieved 97.8% accuracy in recognizing human activities, outperforming a previous SVM-CNN-based method, demonstrating efficiency and improved performance.

Leema [7] and colleagues studied human activity recognition using machine learning. They used a private dataset and they made it using different devices. To analyze human activity, they used a variety of machine classification algorithms, including Random Forest, kNN, Neural Network, Logistic Regression, Stochastic Gradient Descent, and Naïve Bayes. Test accuracy is 99.55%.

Manoj [8] and colleagues studied human activity recognition using PCA and LDA. They used the UCI ML dataset. Results show that LDA is better than PCA. The accuracy of the kernel-SVM has improved from 95.39% to 96.23%. Dimensionality reduction has demonstrated 96.78% accuracy with Naïve Bays. Simple datasets have demonstrated poor accuracy, but dimensionality has enhanced model performance.

2.3. Summary:

This section will give you an idea of all related works or papers like ours paper. Enhancing safety, improving human-robot interaction, and providing tailored monitoring are just a few of the areas in which HAR research has great importance.

Chapter 3: Structure of the system

3. Structure of the system:

3.1. Introduction

Our Human Activity Recognition (HAR) system's design employs cutting-edge machine learning algorithms and wearable sensor technologies to understand and analyze human behavior in a variety of ways. HAR stands for the critical nexus of computer science, engineering, and human-computer interaction. It provides insights on activities in a number of different areas, such as security, fitness, healthcare, and smart surroundings. The requirement for a strong HAR system has grown more and more necessary as wearable gadgets with sensors proliferate. Our system seeks to provide users with tailored insights and useful feedback by utilizing the vast data streams recorded by these sensors to precisely recognize and classify human behaviors in real-time. The work flow is given below:

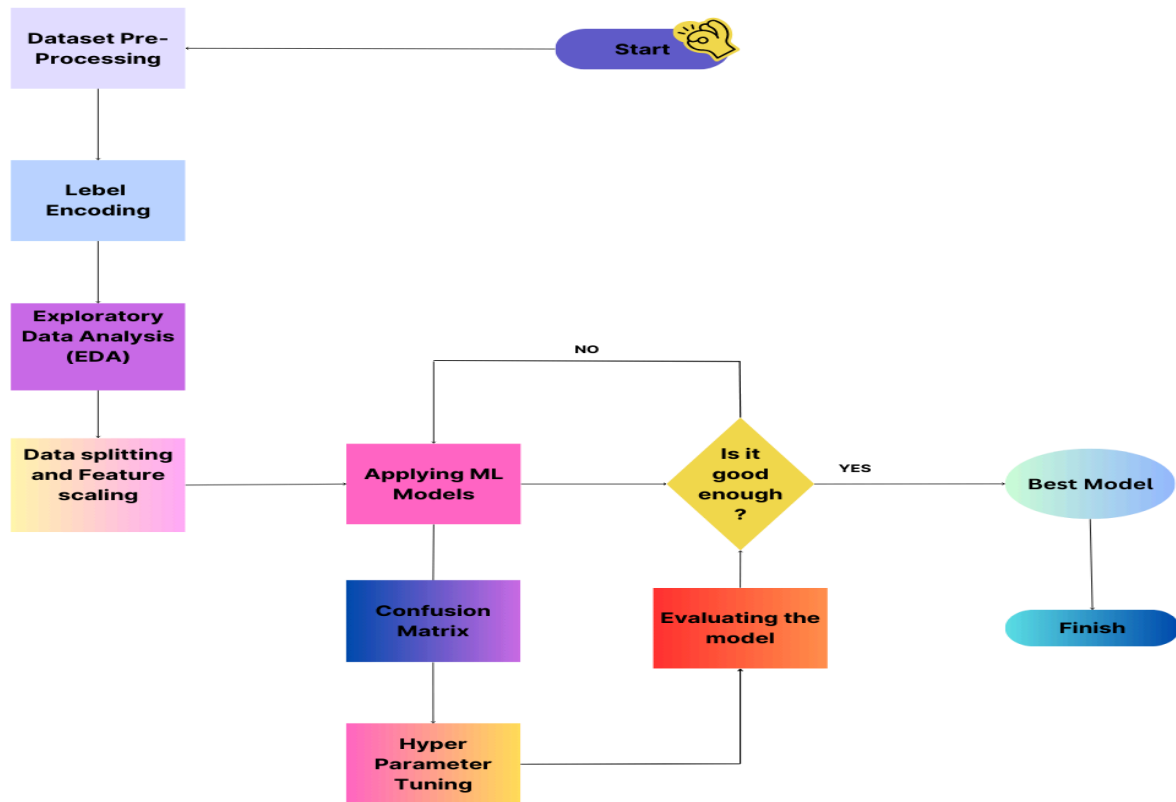


Fig. 1.. Working sequences of the proposed HAR system.

3.2. Dataset

We have used a public dataset from Kaggle. In this dsataset twenty-eight individuals between the ages of 19 and 48 participated in the experiments.

Wearing a smartphone (Samsung Galaxy S II) around their waist, each participant engaged in six different activities: walking, walking upstairs, walking downstairs, sitting, standing, and lying down. We recorded 3-axial angular velocity and 3-axial linear acceleration at a steady rate of 50Hz using its integrated accelerometer and gyroscope. The trials were captured on camera so that the data could be manually labeled. The collected dataset was divided into two sets at random, with 70% of the volunteers chosen to provide training data and 30% to provide test data. Noise filters were used to pre-process the accelerometer and gyroscope sensor signals before sampling them in fixed-width sliding windows with a 50% overlap and a duration of 2.56 seconds (128 readings/window). Using a Butterworth low-pass filter, the sensor acceleration signal—which consists of both body motion and gravitational components—was divided into two categories: body acceleration and gravity. Since it is believed that the gravitational force solely consists of low-frequency components, a filter with a cutoff frequency of 0.3 Hz was employed. A vector of features was derived from each window by computing variables in the frequency and time domains.

3.3. Dataset Preprocessing

Since various machine learning models have different needs and constraints, we use a thorough process in our dataset to prepare the data for analysis. The Decision Tree model is one prominent example of one that requires data preprocessing to ensure robust model performance due to its incapacity to handle null values. We take great care to ensure that our dataset is balanced, full, and free of null or duplicate items. We apply label encoding to translate category variables, such activity kinds, into number equivalents between 0 and 5, which represent the six classes in our dataset, as machine learning models can only process numerical inputs.

Furthermore, we utilize MinMaxScaler for feature scaling to promote uniformity and improve the interpretability of the model by guaranteeing that every feature is within a predetermined range. By reducing the influence of different feature scales, this normalizing phase facilitates more efficient model training and assessment. The preprocessed dataset is then divided into training and testing subsets in an 80:20 ratio to achieve a compromise between performance evaluation and model generalization. The foundation for strong model creation and assessment is laid by this methodical data pretreatment pipeline, which makes it possible to integrate machine learning algorithms for tasks involving the identification of human behavior with ease. Our goal in doing these preparation

steps is to fully use our dataset and optimize the performance of the machine learning models we use for our study.

3.4. Preparing Machine Learning Algorithms

- 3.4.1. GridSearchCV:** Grid search is a parameter tuning technique that systematically builds and examines a model for every set of algorithm parameters given in a grid. It provides the optimal model parameters by fine-tuning the hyperparameter. It improves a model's performance, but it takes a long time because it looks through every possible combination of the specified grid components.
- 3.4.2. RandomizedSearchCV:** In contrast to GridSearchCV, RandomizedSearchCV is a machine-learning method for hyperparameter tuning that selects hyperparameter combinations at random from predetermined ranges. This method preserves the search for the best model configurations while cutting down on computing expenses. RandomizedSearchCV effectively searches hyperparameter spaces by arbitrarily choosing combinations, offering a decent compromise between exploitation and exploration. This approach offers a more useful substitute for thorough grid searches when the search space is large. Through the use of a resource-efficient, randomized search procedure, RandomizedSearchCV improves model performance and aids in the optimization of machine learning models.
- 3.4.3. Explainable AI:** Explainable artificial intelligence refers to a collection of methods and techniques that make it possible for people who are not experts in machine learning to understand and rely on the output of these algorithms. This collection of procedures and techniques is frequently called "Explainable AI." An approach known as "Explainable AI" may be used to explain an AI model, together with its expected effects and any biases. In our approach, we employed LIME, which is incredibly user-friendly.
- 3.4.4. LIME :** "Local Interpretable Model-agnostic Explanations" is what "LIME" stands for in the context of Explainable AI. LIME is a method intended to give comprehensible justifications for the forecasts made by intricate machine learning models, especially when the model's decision-making process might be unclear.

3.5. Machine Learning Model

- 3.5.1. Decision tree:** The decision tree is a widely used machine learning technique because of its capacity to categorize dependent variables into distinct categories. It scans the data and identifies the main independent factors by taking into account several potential variables while generating predictions[11]. The decision tree is a method of supervised categorization. It is a simple structure where judgment outcomes are reflected on terminal nodes and checks on one or more characteristics are displayed on non-terminal nodes. Previous studies on the identification of human activities using decision tree classifiers have produced encouraging results.

- 3.5.2.** Random forest: Random Forest is an ensemble learning method that constructs a multitude of decision trees during training. It combines their outputs through a process known as bagging, where each tree is trained on a random subset of the training data. The final prediction is then determined by a majority vote or averaging. Then each tree produces a prediction, and the most voted prediction is taken as the final output [12]. This approach enhances robustness and generalization, mitigating overfitting. Random Forest is particularly effective for classification and regression tasks, providing high accuracy, handling large datasets with numerous features, and offering feature-importance insights.
- 3.5.3.** Logistic Regression: For applications involving binary classification, a basic machine learning approach is called logistic regression[13]. Despite its name, it is used to estimate the likelihood that an instance will belong to a specific class. The logistic function, which linearly modifies the input information, is used by the algorithm to simulate this probability. Because of its efficiency, interpretability, and simplicity, logistic regression can be used in a variety of contexts. It uses maximum likelihood estimation to optimize coefficients, and predictions are based on a threshold. Logistic regression is a widely used basic technique in the disciplines of finance and medicine for predicting binary outcomes in many machine-learning scenarios.
- 3.5.4.** AdaBoost: AdaBoost, which stands for Adaptive Boosting, is an ensemble learning algorithm for classification and regression [14]. AdaBoost is a learning algorithm that combines weak learners (such as SVC) to form a strong classifier. It is especially effective with weak learners or models that are better than random chance. It constructs a series of weak learners to correct errors and iteratively focuses on misclassified instances, assigning higher weights to them. A weighted average of these learners is used to make the final prediction, resulting in a robust and accurate model.
- 3.5.5.** KNN: K-Nearest Neighbors (KNN) is a simple and universal machine-learning algorithm for classification and regression tasks. It is a type of instance-based learning where the model makes predictions based on the majority class or average of the k-nearest data points in the feature space. "K" is a user-defined parameter representing the number of neighbors to consider. For classification tasks, assign the class label that is most common among the K neighbors to the new data point. For regression tasks, Predict the average or weighted average of the K neighbor's target values. One drawback of KNN is that it can be computationally expensive, especially with large datasets, as it requires calculating distances to all training examples during prediction.
- 3.5.6.** XGBoost: XGBoost is a distributed gradient boosting library that has been optimized to be incredibly effective, versatile, and lightweight[16]. It sequentially builds a series of decision trees, each correcting errors made by the preceding

ones. The model is trained by minimizing a specific objective function, incorporating regularization terms for enhanced predictive performance. XGBoost employs a novel regularization term, the "eXtreme Gradient," that helps control model complexity. It is widely used for classification, regression, and ranking tasks due to its efficiency, speed, and ability to handle large datasets. XGBoost has become a go-to algorithm in machine learning competitions and is known for achieving state-of-the-art results.

3.6. Summary:

The "Structure of the System" section describes our Human Activity Recognition (HAR) system's architecture and setup. The importance of HAR is first explained, then the dataset is described, with special attention to the preparation methods. The preparation of machine learning algorithms, including Explainable AI approaches and hyperparameter tweaking, is then covered in length in this part. It concludes by summarizing the system's design and methodology and offering a plan for further development and assessment of the HAR system.

Chapter 4: Technical Details

4. Technical Details

4.1.Introduction:

The increasing prevalence of wearable devices has paved the way for innovative applications in human activity recognition (HAR), contributing to fields such as healthcare, sports, and lifestyle monitoring. This report aims to provide a comprehensive comparative analysis of various machine learning (ML) models used for HAR, specifically focusing on wearable devices.

4.2. Existing Solution:

Currently, many HAR systems employ traditional machine learning algorithms and deep learning techniques to interpret sensor data collected from wearables. Commonly used models include Decision Trees, Support Vector Machines (SVM), Random Forests, and Deep Neural Networks (DNN). These models exhibit varying levels of accuracy, computational efficiency, and robustness in different scenarios.

4.3. Proposed Solution:

Our proposed solution involves evaluating and comparing the performance of several ML models in the context of wearable-centric HAR. We will focus on the following models:

- 4.3.1. **Decision Trees:** A tree-based model known for its interpretability and ease of use.
- 4.3.2. **Support Vector Machines:** Effective in high-dimensional spaces, particularly useful for complex activity recognition tasks.
- 4.3.3. **Random Forests:** An ensemble learning method that combines multiple decision trees for improved accuracy and generalization.
- 4.3.4. **Deep Neural Networks:** Specifically, convolutional neural networks (CNNs) and recurrent neural networks (RNNs), which have shown success in feature extraction and temporal modeling.

The evaluation criteria include accuracy, precision, recall, F1 score, computational efficiency, and scalability.

4.4. Design Alternatives:

We consider various design alternatives to enhance the performance of the selected models:

- 4.4.1. **Feature Engineering:** Experimenting with different feature sets extracted from sensor data to improve model interpretability and generalization.
- 4.4.2. **Hyperparameter Tuning:** Optimizing model parameters to achieve the best balance between bias and variance, improving overall performance.
- 4.4.3. **Transfer Learning:** Exploring the feasibility of transferring knowledge from pre-trained models to accelerate training and enhance performance on specific tasks.
- 4.4.4. **Ensemble Methods:** Investigating the use of ensemble methods to combine predictions from multiple models, potentially improving accuracy and robustness.

4.5. Summary:

In conclusion, this report aims to provide a thorough analysis of ML models for wearable-centric HAR. By comparing the performance of Decision Trees, Support Vector Machines, Random Forests, and Deep Neural Networks, we seek to identify the most suitable model for accurate and efficient activity recognition. Through the exploration of design alternatives, we aim to enhance the overall performance of the selected models, ultimately contributing valuable insights to the evolving landscape of wearable device applications.

Chapter 5: Essential Parts

5. Essential Parts

5.1. Introduction

Human activity recognition (HAR) using wearable devices has gained significant attention in recent years due to the increasing prevalence of wearable technology and the potential applications in healthcare, fitness tracking, and lifestyle monitoring. This chapter provides an overview of the essential components and considerations in building a machine learning (ML) model for HAR using wearable devices.

The objective of this project is to conduct a comparative analysis of different ML models for accurate and efficient human activity recognition based on data collected from wearable devices. By understanding the essential parts of the project, we aim to highlight the key aspects that contribute to the success of the proposed models.

5.2. Essentials for Building This Project

To build an effective human activity recognition system for wearable devices, several essential components must be considered:

5.2.1. Data Collection:

5.2.1.1. Selection of Wearable Devices: Choose appropriate wearable devices capable of collecting relevant sensor data, such as accelerometers and gyroscopes.

5.2.1.2. Data Preprocessing: Clean and preprocess the collected data to remove noise, outliers, and ensure consistency across different devices.

5.2.2. Feature Extraction:

5.2.2.1. Identify Relevant Features: Extract meaningful features from the sensor data that capture essential patterns related to human activities.

5.2.2.2. Dimensionality Reduction: Apply techniques to reduce the dimensionality of the feature space, enhancing computational efficiency and model performance.

5.2.3. Model Selection:

5.2.3.1. Comparative Analysis: Evaluate and select ML models suitable for HAR, considering factors like accuracy, computational efficiency, and interpretability.

5.2.3.2. Deep Learning vs. Traditional ML: Compare the performance of deep learning models (e.g., neural networks) with traditional ML algorithms (e.g., decision trees, support vector machines).

5.2.4. Model Training and Validation:

5.2.4.1. Splitting the Dataset: Divide the dataset into training and validation sets to train and evaluate model performance.

- 5.2.4.2. **Hyperparameter Tuning:** Optimize model hyperparameters to enhance the overall performance and generalization capabilities.

5.3. Test Equipment

To conduct a comprehensive analysis, the following test equipment will be used:

- 5.3.1. **Wearable Devices:** Select a range of popular wearable devices to collect diverse data for testing and validation.
- 5.3.2. **Computing Resources:** Utilize high-performance computing resources for training complex ML models efficiently.
- 5.3.3. **Data Visualization Tools:** Implement tools for visualizing sensor data, model predictions, and performance metrics to facilitate analysis.

5.4. Block Diagram

The block diagram provides a visual representation of the key components in the human activity recognition system. It outlines the flow of data from wearable devices through preprocessing, feature extraction, model training, and validation stages. The block diagram serves as a reference for understanding the overall architecture of the proposed system.

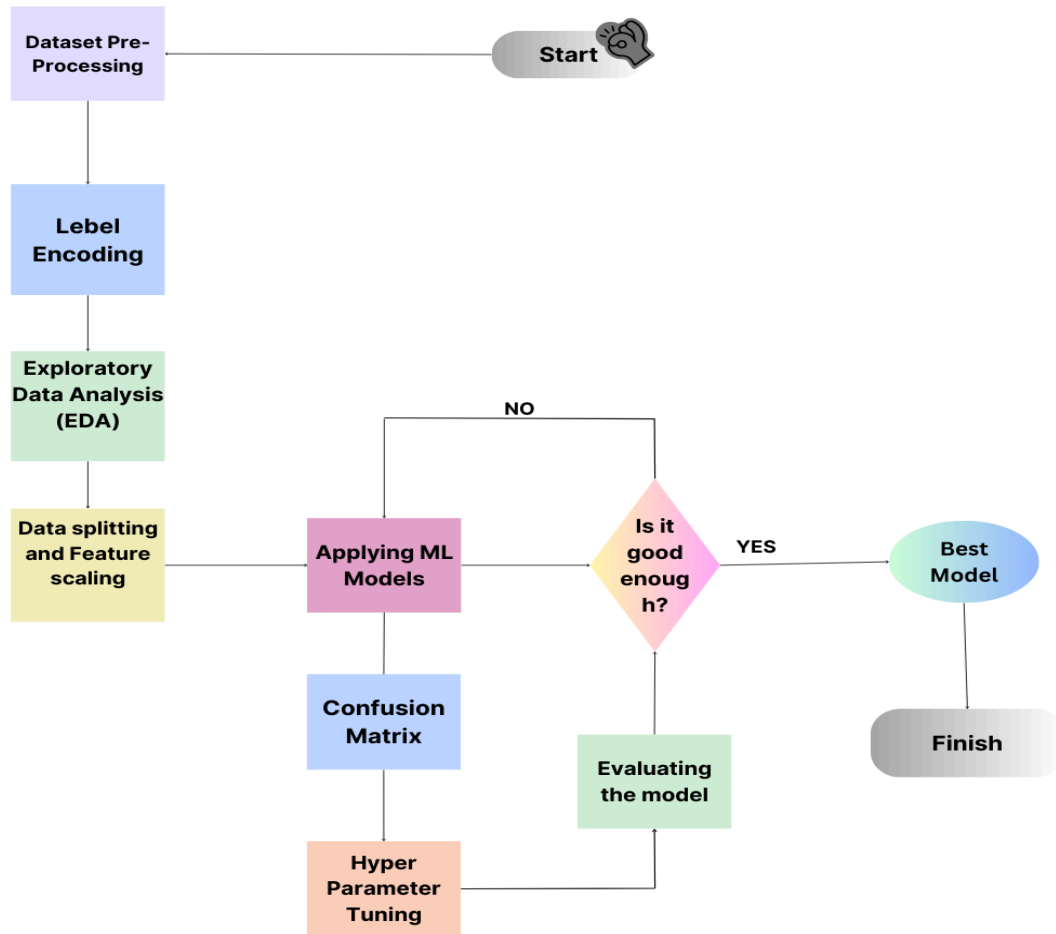


Fig. 2. Working sequences of the proposed HAR system.

5.5.Summary

In summary, this chapter has outlined the essential components for building a human activity recognition system for wearable devices using machine learning models. By carefully considering data collection, feature extraction, model selection, and testing equipment, we aim to create a robust framework for comparative analysis. The subsequent chapters will delve into the specific methodologies, results, and discussions of the comparative analysis, providing valuable insights into the effectiveness of different ML models for wearable device-centric human activity recognition.

Chapter 6: Compliance with standards

6: Compliance with standards

6.1: Introduction

For the machine learning project "A Comparative Analysis of Different ML Models for Wearable Devices Centric Human Activity Recognition" to be successful, dependable, and secure, standards compliance is crucial. The project complies with a number of industry norms and regulations to provide a reliable and smooth customer experience.

6.2: ML Model Development Standards

The project includes several steps, from data preparation to model validation, and adheres to strict guidelines for the construction of machine learning models. The research makes use of well-established techniques like hyperparameter tuning and cross-validation to guarantee the models' generalization and dependability over a variety of datasets and real-world situations.

6.3: Data Privacy and Security Standards

Throughout the course of the project, strict procedures are put in place to protect the security and privacy of user data. It is assured that data protection laws, such as GDPR and HIPAA, are followed, and risks of unauthorized access or data breaches are reduced by using encryption and access restrictions.

6.4: Ethical Considerations

The creation and use of ML models are guided by ethical guidelines that address issues with transparency, bias reduction, and fairness. The project places a high priority on algorithmic openness, which helps to build confidence and accountability by allowing stakeholders to comprehend how the models make decisions.

6.5: Interoperability Standards

The project conforms to industry standards for compatibility requirements, communication protocols, and data exchange formats, ensuring smooth interaction with wearables and other pertinent technology. This makes it simple to integrate wearables with current technology and makes cross-platform data exchange and collaboration easier.

6.6: Model Interpretability and Explainability

Making sure ML models are interpretable and explainable is emphasized so that stakeholders can understand the reasoning behind the predictions and choices made by the models. To improve transparency and foster user confidence, methods like feature significance analysis and model visualization are used.

6.7: Continuous Monitoring and Evaluation

The effectiveness and dependability of ML models in practical contexts are evaluated through the implementation of procedures for ongoing monitoring and assessment. Consistent audits and feedback loops provide incremental enhancements and guarantee the models' long-term efficacy in precisely identifying human actions from wearable device data.

6.8: Documentation and Reporting Standards

Throughout the project lifespan, thorough documentation is kept up to date, including thorough explanations of data sources, preprocessing procedures, model designs, and assessment outcomes. Transparency and accountability are ensured by the research community's capacity to share information and reproduce findings when clear and succinct reporting guidelines are followed.

6.9: Summary

The development and implementation of ML models for wearable device-centric human activity recognition is ensured to be ethical, trustworthy, and transparent through the careful adherence to industry standards and guidelines across a number of project facets, including ML model development, data privacy and security, ethical considerations, interoperability, model interpretability, continuous monitoring and evaluation, documentation, and reporting. This dedication promotes innovation and growth in the wearable and machine learning fields while improving user happiness, trust, and well-being.

Chapter 7: Design Impact

Chapter 7: Design Impact

7.1: Introduction

"Human activity recognition" (HAR) is a subfield of computer science and engineering that uses wearable technology to automatically recognise and categorize human behaviors. HAR analyzes human motion, including gestures, movements, and behaviors captured by sensors, cellphones, or photographs. It does this by utilizing techniques like computer vision and sensor technology. Artificial agents must be capable of identifying complex environmental signals in order for people and robots to collaborate in a variety of scenarios. For this reason, human-robot interaction (HAR) is essential. The impact of Human Activity Recognition projects is examined in depth in this chapter, which summarizes the consequences from a variety of angles including economic, social, political, ethical, environmental, health, safety, and manufacturability.

7.2: Project Prototype

1. Challenges in ML Model Development:

- **Model Adaptability:** It might be difficult to guarantee that machine learning models can correctly identify a wide range of human activities in various situations and user behaviors.
- **Data Variability and Labeling:** Careful consideration must be given throughout model building to curate annotated datasets that accurately replicate real-world events and to ensure label correctness.
- **Trade-off Between Complexity and Performance:** For smooth integration into wearable devices with constrained processing power, a balance between model complexity and computational efficiency is necessary.

2. Cost-Effectiveness in Model Deployment:

- **Leveraging Pre-trained Models:** Development costs may be drastically lowered and deployment durations can be accelerated by making use of pre-trained machine learning models and transfer learning strategies.

- **Edge Computing Solutions:** By putting lightweight machine learning models directly on wearables, you can reduce the amount of time and money you spend on cloud infrastructure.

3. Model Unit Cost Analysis:

- **Development Costs:** These comprise charges for gathering data, training models, and refining algorithms.

- **Deployment Costs:** Comprise costs related to user interface design, software development, and hardware integration.

- **Maintenance & Upkeep:** Take into consideration recurring expenses for bug patches, model upgrades, and user support services.

4. User Acceptance and Adoption:

- **Intuitive User Interfaces:** Increasing user engagement and adoption rates is possible via the design of user-friendly interfaces and the provision of useful insights.

- **Performance vs. Privacy Trade-offs:** Building trust and acceptance requires striking a balance between the requirement for precise activity detection and user privacy preferences and data security concerns.

5. Overcoming Difficulties Technically:

- **Methods of Sensor Fusion:** Strengthening and improving the accuracy of activity identification is achieved by combining data from several sensors, including gyroscopes and accelerometers.

- **Model Optimization Strategies:** By lowering model size and inference latency, methods like pruning, quantization, and knowledge distillation can be used to enable real-time deployment on wearables.

7.3: Economic Impact

7.3.1: Efficiency and Cost Cutting

- **Savings on healthcare costs:** Precise HAR models provide proactive health monitoring and early intervention, which may lower medical expenses related to avoidable illnesses.

- **Productivity Gains:** Wearables with ML-powered HAR capabilities make workdays easier and boost worker productivity, which has a positive impact on enterprises' bottom lines.

7.3.2: Market Penetration and Revenue Generation:

- **Market Differentiation:** By providing wearables with cutting-edge HAR features, brands stand out in a competitive market, drawing in discriminating customers and boosting sales.

- **Subscription-Based Models:** Recurring income streams and long-term profitability are ensured by monetizing premium services like activity analytics and personalised coaching through subscription plans.

7.4: Social Impact

7.4.1: Health and Well-being

- **Chronic Disease Management:** By enabling continuous monitoring of actions pertinent to managing chronic diseases, ML-enabled HAR empowers people to take proactive responsibility for their health.

- **Behavior Modification:** Wearable technology's real-time input encourages users to follow recommended physical activity levels and adopt healthier lifestyles, which improves their general wellbeing.

7.4.2: Accessibility and Inclusivity

- **Inclusive Design Practices:** To ensure inclusiveness and accessibility for users from a variety of backgrounds, machine learning models must be trained on varied demographic datasets and take contextual and cultural aspects into consideration.
- **Assistive Technologies:** For people with mobility disabilities, wearables with HAR capabilities are a useful assistive technology that can improve their autonomy and quality of life.

7.5: Environmental Impact

7.5.1: Sustainable Living

- **Energy-Efficient Design:** By optimizing machine learning algorithms for low power consumption, wearable devices' batteries last longer, resulting in less technological waste and energy usage.
- **Encouragement of Walking and Bicycling as Viable forms of Transportation:** HAR enabled wearables support sustainable urban mobility and lower carbon emissions by encouraging walking and cycling as viable forms of transportation.

7.5.2: Green Technology Adoption

- **Eco-Friendly Materials:** Using sustainable materials and manufacturing techniques when designing wearables helps to lessen the environmental impact of electronics while also supporting environmental conservation aims.
- **End-of-Life Recycling:** Encouraging appropriate disposal methods and including recyclable parts guarantees the environmentally beneficial lifetime management of wearable technology.

7.6: Ethical Impact

7.6.1: Data Privacy and Security

- **User Consent and Transparency:** Wearable device collection of sensitive personal data is protected by upholding user privacy rights through explicit consent methods and clear data processing policies.

- **Secure Data Transmission:** Encryption methods and secure communication routes shield user data from malevolent assaults and unauthorized access, maintaining data confidentiality and integrity.

7.6.2: Fairness and Bias Mitigation

- **Bias detection and mitigation:** To reduce the possibility of discriminating results and guarantee equal HAR across a range of user demographics, fairness-aware algorithms should be implemented and ML models should be routinely audited for biases.

- **Ethical Standards Compliance:** Users and stakeholders are more likely to trust a wearable device developed in accordance with ethical principles when regulations and ethical guidelines are followed during data collection, model training, and deployment.

7.7: Health and Safety Impacts

7.7.1: Injury Prevention

- **Fall Detection:** The ability to identify falls in elderly and vulnerable populations is made possible by ML-powered HAR algorithms, which also reduce the probability of injuries occurring.

- **Activity Monitoring:** By enabling prompt treatments and enhancing overall health outcomes, continuous activity monitoring helps identify aberrant behavior patterns that may indicate a health emergency.

7.7.2: Public Health Surveillance

- **Disease Outbreak Detection:** Wearable activity data aggregates are a powerful epidemiological indicator that supports public health monitoring efforts by helping to detect and contain infectious disease outbreaks early on.

Population Health Insights: By analyzing HAR data at scale, public health policies and initiatives targeted at disease prevention and health promotion may be informed by insights into health trends and activity patterns at the population level.

7.8: Manufacturability

7.8.1: Scalable Production

- **Standardized Components:** Creating wearables with modular parts and standardised interfaces expedites manufacturing and permits quick scalability to satisfy consumer demand.

- **Just-in-Time Manufacturing:** By using agile manufacturing techniques and streamlining the supply chain, inventory costs are reduced and production efficiency is increased, guaranteeing that wearables are delivered to customers on schedule.

7.8.2: Quality Assurance and Testing

- **Sturdy Testing Frameworks:** Under a range of operating situations, wearable devices' dependability, robustness, and safety are guaranteed by conducting thorough quality assurance tests and validation processes.

- **User-Centric Design Iterations:** By asking end users for input and implementing design changes based on field tests and usability studies, you may increase product acceptability and user happiness, which promotes ongoing innovation and market relevance.

7.9: Sustainability of the Project

- **Diversified money Sources:** Long-term project viability is maintained and financial resources are diversified when money is obtained through government grants, commercial investors, and research collaborations.
- **Ongoing R&D Initiatives:** Funding these projects keeps the project at the forefront of wearable technology breakthroughs and encourages technical innovation.
- **Standards and Regulatory Conformity Adherence:** Maintaining industry standards for data privacy, security, and product safety as well as regulatory compliance build the project's reputation and consumers' and regulators' confidence.

7.10: Summary

The project "A Comparative Analysis of Different ML Models for Wearable Devices Centric Human Activity Recognition" presents a comprehensive exploration of machine learning (ML) models tailored for human activity recognition (HAR) within wearable technology, addressing technological challenges, economic implications, social considerations, environmental factors, ethical imperatives, and health impacts. Through innovative solutions such as sensor fusion techniques and model optimization strategies, the project navigates challenges while emphasizing economic benefits, societal empowerment, environmental conservation, ethical principles, and health advancements. By prioritizing scalable production, quality assurance, regulatory compliance, and diversified funding, the project ensures sustainability while driving forward the transformative potential of ML-driven HAR in reshaping health monitoring, environmental sustainability, and user experiences within wearable technology.

Chapter 8: Work Sheets

Chapter 8: Worksheet

8.1: Introduction

We set out to comprehend and use machine learning algorithms designed to analyze human behaviors recorded by wearable sensors in our research of Human Activity Recognition (HAR). With the goal of using complex algorithms to automate the detection and classification of human actions, HAR was a crucial topic that connected computer science and engineering. Through the utilization of sensor and computer vision technology, HAR was able to give insights on human behavior, gestures, and motion.

We examined actions, gestures, and activities captured by sensors that embodied the core of human activity. With HAR, we aimed to precisely recognise and classify human behaviors by using machine learning models to analyze data from wearables such as cameras, sensors, and smartphones. We were able to better understand complex behaviors by combining machine learning techniques such as AdaBoost, XGBoost, KNN, Random Forest, Decision Tree, and Logistic Regression and exploring the spatiotemporal context of human movements in detail.

8.2: Project Initiation

8.2.1: Defined Project Scope

The scope of our study included a thorough investigation of HAR approaches, with an emphasis on the creation and assessment of machine learning models for the identification of human activities. Our goal was to lay out precise goals and benchmarks that would direct us as we implemented and evaluated different algorithms.

8.2.2: Identified Stakeholders

Researchers, data scientists, engineers, and those working on the creation and implementation of machine learning models for HAR were important project participants. Important stakeholders included healthcare providers, exercise instructors, and security staff, who were also users and benefactors of HAR technology.

8.2.3: Established Project Objectives

One of our main goals was to investigate the most recent machine learning methods available for HAR.

2. Applying and optimizing machine learning models to reliably identify activities.
3. Assessing various algorithms' performance with reference datasets.
4. Improving comprehension of HAR applications across a range of industries, including as industrial safety, security, fitness, and healthcare.

8.3: Planning and Design

We built the groundwork for our project during the planning and design phase by describing the techniques, resources, and equipment needed to put machine learning models into practice. To guarantee the efficacy of our solutions, we gave robustness, scalability, and interpretability top priority in our methodology.

8.4: Development

We built the groundwork for our project during the planning and design phase by describing the techniques, resources, and equipment needed to put machine learning models into practice. To guarantee the efficacy of our solutions, we gave robustness, scalability, and interpretability top priority in our methodology.

8.5: Testing

User Acceptance Testing

Feedback from stakeholders and end users was gathered during user acceptability testing to evaluate the accuracy, dependability, and usefulness of our HAR models. To increase overall pleasure and efficacy, we made design iterations based on user feedback.

8.6: Deployment

We deployed our HAR models in real-world applications after testing and validation went well, guaranteeing a smooth integration with current workflows and systems. In order to encourage end users to utilize HAR technology, we offered them assistance and training.

8.7: Documentation

8.7.1: Technical Documentation

Our HAR models' implementation details, algorithms, and methodology were all thoroughly explained in the technical documentation. It functioned as a guide for researchers and developers who wanted to expand or duplicate our work.

8.7.2: User Manuals

Our HAR systems' user manuals included thorough instructions on how to operate and understand their outputs. Their goal was to provide end users with the information and abilities needed to properly utilize HAR technology.

8.7.3: Training Materials

Training resources helped stakeholders and end users enhance their skills and convey knowledge. They comprised instructional materials, workshops, and tutorials aimed at improving comprehension and skill with HAR applications and ideas.

8.8: Financial Plan for Project

8.8.1: Estimated Costs

Development Costs: Including setup fees for infrastructure, data collection, and software development.

Testing Costs: Included resources allotted for quality control, validation datasets, and testing settings.

Deployment Costs: Included were integration, tools for deployment, and user training related to the implementation of HAR models in production settings.

Maintenance Cost: Allocate funds for continuous upkeep, upgrades, and assistance services to guarantee the HAR systems' long-term viability.

8.8.2: Revenue Projections

The prospective revenue streams or returns on investment from the implementation and use of HAR technologies were shown in revenue predictions. Revenue sources from product sales, collaborations, licensing, and subscription services might have been among them.

8.9: Feasibility Study for Parking Chai

Technical Feasibility

The potential of HAR technology implementation within the limitations of infrastructure, resources, and technological capabilities was evaluated in terms of technical feasibility. It entailed assessing the necessary hardware and software, the accessibility of the data, and the system's compatibility.

8.10: Summary

In conclusion, the construction and assessment of machine learning models was the primary goal of our study, which sought to progress the field of human activity recognition. We aimed to uncover wearable device potential for greater security, improved industrial safety, personalized healthcare monitoring, and adaptable smart environments by utilising cutting-edge algorithms and approaches. By means of painstaking planning, development, and testing, we made an effort to provide HAR solutions that were reliable and efficient, catering to the various requirements of stakeholders and end users.

Chapter 9: Result and Discussion

9. Result and Discussion

9.1. Introduction:

This section of the paper discusses the results of all different Machine Learning algorithms for Human Activity Recognition. We have applied six different Machine Learning models in the dataset "Human Activity Recognition with Smartphones" from Kaggle. In our dataset, there are 563 columns and 7352 rows of attributes. we splitted the data into train (80%) and test (20%) data. In 563 columns, we used one column as our target variable column, which has 6 classes which is a multiclass problem, and the other 562 columns are features or predictors. We evaluate the performance metrics from the model, including accuracy, Recall, F1 score, and precision. These matrices can be calculated through the following equations:

$$\text{Accuracy} = (\text{correct predictions}) / (\text{all predictions}) = (TP + TN) / (TP + TN + FP + FN) \quad [1]$$

$$\text{Precision}_c = TP_c / (TP_c + FP_c) \quad [2]$$

$$\text{Recall}_c = TP_c / (TP_c + FN_c) \quad [3]$$

$$\text{F1_Score}_c = 2 * (\text{Precision}_c * \text{Recall}_c) / (\text{Precision}_c + \text{Recall}_c) \quad [4]$$

Where TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives, c = class label (1 to 6), TP_c = True Positives for class c, FN_c = False Negatives for class c, FP_c = False Positives for class c, Precision_c = Precision for class c, Recall_c = Recall for class c, F1 Score_c = F1 score for class c. We calculate all the values above the confusion matrix for our six classes.

		Predicted Class					
		Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Actual Class	Class 1	TN	TN	FP	TN	TN	TN
	Class 2	TN	TN	FP	TN	TN	TN
	Class 3	FN	FN	TP	FN	FN	FN
	Class 4	TN	TN	FP	TN	TN	TN
	Class 5	TN	TN	FP	TN	TN	TN
	Class 6	TN	TN	FP	TN	TN	TN

Fig. 3. Confusion matrices for class 3.

Figure 3, shows the confusion matrices for class number 3. Following this we calculate accuracy, recall, F1 score, and precision. for our other classes.

9.2 Summary

TABLE II. HYPERPARAMETER VALUES' RANGES FOR VARIOUS ML MODELS

Sl. No.	Model	Hyperparameter Value Range	Optimized value
1.	XG Boost	n_estimators: [100, 200, 300], max_depth: [3, 4, 5], learning_rate: [0.01, 0.1, 0.2], subsample: [0.7, 0.8, 0.9], colsample_bytree: [0.7, 0.8, 0.9], objective: ['binary:logistic', 'multi:softmax'],	subsample: 0.9, objective: 'multi:softmax', n_estimators: 300, max_depth: 5, learning_rate: 0.1, colsample_bytree: 0.9
2.	Random Forest	n_estimators': [100, 200, 300], max_depth: [None, 10, 20, 30], min_samples_split: [2, 5, 10], min_samples_leaf: [1, 2, 4], criterion: ['gini', 'entropy']	n_estimator': 200, min_samples_split: 5, min_samples_leaf: 1, max_depth: None, criterion: 'entropy'

3.	Logistic Regressi on	'C': [0.001, 0.01, 0.1, 1, 10], 'penalty': ['l1', 'l2'], 'solver': ['liblinear', 'newton-cg', 'lbfgs', 'saga']	'C': 10, 'penalty': 'l1' 'solver': 'liblinear'
4.	Decision Tree	max_features: [auto, sqrt, log2], max_depth: np.linspace(20, 1000, 50), min_samples_split: [2, 5, 10, 14], min_samples_leaf: [1, 2, 4, 6, 8], criterion: [entropy, gini]	max_features: [sqrt], max_depth: 520, min_samples_split: 10, min_samples_leaf: 4, criterion: entropy
5.	KNN	p: [1,2], n_neighbors: [1-30], leaf_size: [1-50]	p: 1, n_neighbors: 1, leaf_size: 1
6.	AdaBoo st	n_estimators : [50, 200, 500], Learning_rate: [0.01, 0.1, 1.0],	n_estimators: 200, learning_rate: 0.01

Table II illustrates the hyperparameter values' ranges for all the ML models.

TABLE III. PERFORMANCE METRICS OF VARIOUS CLASSIFIERS FOR DEFAULT PARAMETERS

Classifier	Precision	Recall	F1 Score	Accuracy
------------	-----------	--------	----------	----------

XG Boost	99%	99%	99%	99.39%
Random Forest	98%	98%	98%	98.23%
Logistic Regression	99%	99%	99%	99%
Decision Tree	94%	94%	94%	94%
KNN	96%	96%	96%	95.93%
AdaBoost	88%	80%	75%	80.16%

Table III illustrates the performance matrices of various classifiers we used for default parameters. As per this table, XGBoost outperforms all the machine learning models with the highest accuracy of 99.39 percent and Decision Tree has the lowest accuracy of 80.16 percent.

TABLE IV. PERFORMANCE METRICS OF VARIOUS CLASSIFIERS WITH OPTIMIZED HYPERPARAMETERS

Classifier	Precision	Recall	F1 Score	Accuracy
XG Boost	100%	100%	100%	99.52%
Random Forest	99%	99%	99%	98.64%

Logistic Regression	99%	99%	99%	99%
Decision Tree	94%	94%	94%	93.55%
KNN	98%	98%	98%	98.31%
AdaBoost	97%	97%	97%	96.61%

TABLE IV, we see performance metrics of various classifiers with optimized hyperparameters . Here eight models are used. As per this table, XGBoost outperforms all the machine learning models with the highest accuracy of 99.52 percent and Decision Tree has the lowest accuracy of 93.55 percent.

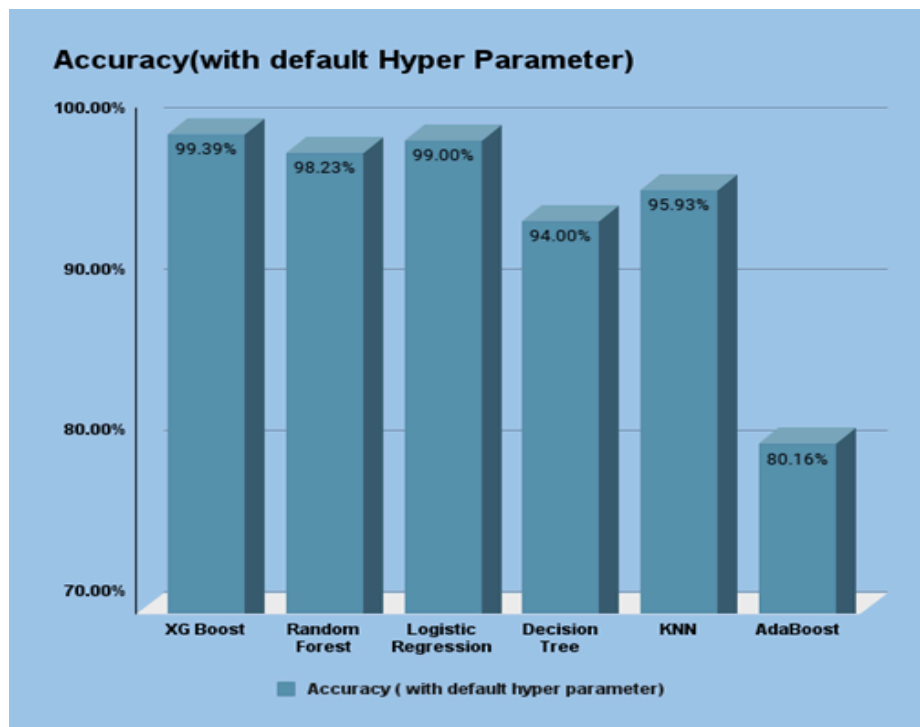


Fig. 4: Accuracy of all ML models with default hyperparameters.

Fig. 4.1 shows the accuracy of the ML models in the form of a bar graph with the default parameters technique. It shows that XGBoost has the best accuracy of 99.39% for the dataset and AdaBoost has the lowest accuracy of 80.16%.

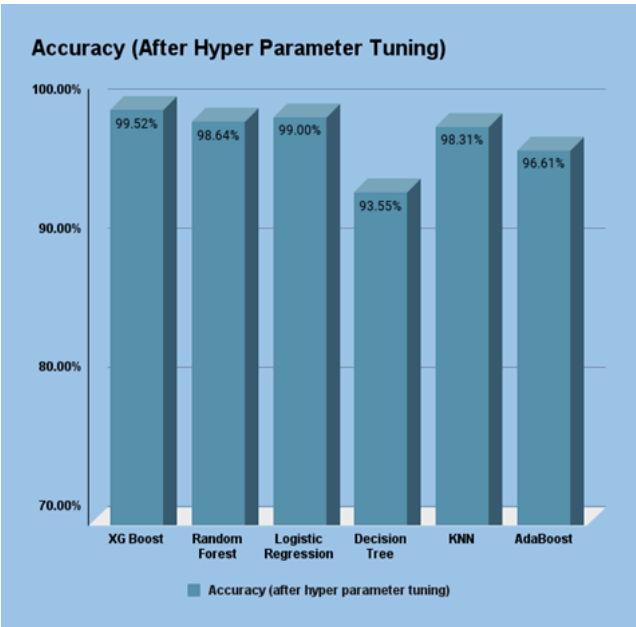


Fig. 5: Accuracy of all ML models with optimized hyperparameters using GridSearchCV or RandomSearchCV

Fig. 4.2 shows the accuracy of the ML models in the form of a bar graph with optimized hyperparameters using GridSearchCV or RandomSearchCV. It shows that XGBoost has the best accuracy of 99.52% for the dataset and Decision Tree has the lowest accuracy of 93.55%.

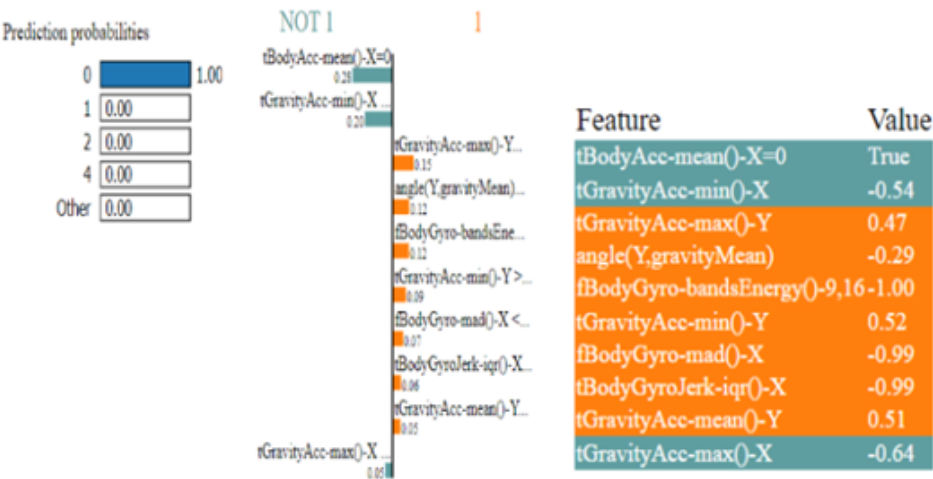


Fig. 6 Machine learning model prediction interpretation by LIME explainable AI library

In Fig-5, Using the LIME explainable AI framework, the logistic regression model's prediction interpretation is displayed in Fig.5. Figure 4 shows that, after hyperparameter optimization, the

XG Boost model performs the best. The LIME prediction results are produced using this machine-learning model. The XG Boost model predicted human activity with an accuracy of 99.52%, as shown by the LIME figure. Based on tGravityAcc-max()-Y of 15% , angle(Y,gravityMean) of 12%, fBodyGyro-bandsEne 12%, tGravityAcc-min()-Y greater than 9%, fBodyGyro-mad()-X less than 7%, tBodyGyroJerk-iqr()-X 6%, tGravityAcc-mean()-Y 5%, the human activities are predicted for everyone when the value of tBodyAcc-mean()-X=0, tGravityAcc-min()-X, tGravityAcc-max()-X are True, -54%, -64% respectively.

TABLE V. Comparison between other papers and our works

Refe renc es	Dataset	ML model	Accuracy	Recall
[3]	Private	K-mean and ANN	98.24%	N/A
[4]	MHEALTH	RNN(GPU, SRU)	99.80%	N/A
[5]	NTU RGB+D, SBU Interaction, UT-Kinect, Berkeley MHAD	HBNN	100%	N/A
[6]	UNI-HAR	ResNet	97.6%	N/A
[7]	Public	Feed-forward, KNNs,SVM, Discriminant analysis, and Naive Bayes	98%	100%
[8]	Public (PPG data collected	Random Forest	97.8%	N/A

	from the physionet)			
[9]	Private	R.F, N.V	99.55%	N/A
[10]	Public (UCI ML)	SVM, Naive Bayes	96.78%	N/A
This work	Human Activity Recognizatio n Using SmartPhone	DT,RF ,LR, KNN, XG Boost, AdaBoost	99.52%	100%

Table V In this table, we see the existing work on the Human Recognition System. We describe various papers and show dataset type, ml model, accuracy, and recall. In reference 3, they use a private dataset, k-mean and ANN model. The accuracy is 98.24%. In reference 4, they use MHEALTH dataset, RNN(GPU,SRU) model. The accuracy is 99.80%. In reference 5, they use NTURGB+D dataset, HBRNN model. The accuracy is 100%. We also describe five more papers details. Then compare all the criteria with this work. Here we use Human Activity Recognition on UsingSmartPhone dataset and DT,RF, LR,KNN, XGBOOST and AdaBoost Model. The accuracy is 99.52% and recall value 100.

Chapter 10: Future Work

10. Future Work

10.1 Introduction

Due to its many uses in human-computer interaction, fitness tracking, and healthcare, Human Activity Recognition (HAR) using smartphone data has attracted a lot of attention. Using information gathered from smartphones, we used a variety of machine learning algorithms in this work to categorize human behaviors. Our findings demonstrate how well various models distinguish human actions, with XGBoost appearing as the best model following hyperparameter adjustment.

10.2 Summary

Feature Engineering: Investigate cutting-edge methods for feature engineering to extract more insightful characteristics from sensor data from smartphones. To improve model performance, this could entail frequency domain modifications, time-series analysis, or feature selection techniques.

Algorithm Ensemble: To further increase classification accuracy, combine several machine learning algorithms to explore the possibilities of ensemble approaches. To maximize each model's strengths and minimize its faults, strategies like stacking or mixing could be used.

Deep Learning Approaches: Using smartphone data, investigate the suitability of deep learning models for HAR, such as convolutional neural networks (CNNs) or recurrent neural networks (RNNs). Deep learning architectures may provide improved performance for complicated activity recognition, as they have demonstrated encouraging results in comparable challenges.

Online Learning: Put online learning techniques into practice to allow for ongoing model update and modification in response to changing activity patterns. Because of this, real-time HAR applications would be made easier, allowing the model to adapt dynamically to changing data streams.

Cross-Domain Generalization: Assess how well the trained models generalize to various smartphone datasets and activity scenarios. In order to evaluate the models' resilience and suitability for a range of situations, testing them on various datasets would be necessary.

Interpretability and Explainability: To improve model transparency and offer insights into forecasts, use interpretability approaches like LIME or SHAP. It is essential to comprehend the reasoning behind model decisions, particularly in important fields like healthcare.

Deployment and Integration: Create systems or apps that are easy to use for the practical implementation of HAR solutions. Widespread adoption and useful application of the created models would be made possible by integration with wearable technology or smartphone applications.

Validation and Evaluation: Using separate datasets and in-person experiments, thoroughly validate and assess the suggested models. Thorough testing in a variety of settings would confirm the HAR system's dependability and efficiency.

Chapter 11: Conclusion

11. Conclusion

11.1 Introduction

In summary, this research delves into the realm of Human Activity Recognition (HAR) through an extensive exploration of machine learning models applied to data from wearable devices. The study employs a diverse set of classifiers, including XG Boost, Random Forest, Logistic Regression, Decision Tree, KNN, and AdaBoost, assessing their performance under both default and hyperparameter-tuned settings. The results showcase XG Boost as the consistently superior performer, emphasizing its adaptability and resilience. However, the study also highlights the nuanced strengths of other models, such as the robustness of Logistic Regression and Random Forest, and the competitive accuracy of Decision Tree and KNN. The significance of hyperparameter tuning is underscored, with notable improvements observed across models, particularly in the case of AdaBoost.

11.2 Conclusion

Moreover, the research integrates Explainable AI, utilizing LIME to interpret the predictions of the logistic regression model. This emphasis on interpretability contributes to the broader understanding of how machine learning models arrive at their predictions. The study concludes by emphasizing the need for tailored approaches in configuring models, acknowledging the specific nuances of datasets, and recognizing the ongoing refinement required for optimal model performance. Overall, this work not only advances the understanding of HAR but also provides valuable insights into the intricate interplay between machine learning models, hyperparameter tuning, and interpretability in the context of wearable device-centric human activity recognition. In our research on "A Comparative Analysis of Different ML Models for Wearable Devices Centric Human Activity Recognition," we plan to refine feature engineering techniques for extracting relevant information from sensor data. Exploring advanced model architectures, including recurrent neural networks (RNNs), will enhance recognition accuracy. We aim to conduct an extensive hyperparameter search and dataset expansion for a comprehensive evaluation of model generalization. Integrating user feedback and ensuring ethical considerations guide our future work for responsible and personalized deployment of activity recognition models.

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Code:

```
#ignore warnings
import warnings
warnings.filterwarnings('ignore')

#standard Libraries
import pandas as pd
import numpy as np

#visualization
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
plt.rcParams['figure.figsize'] = (10,8)
```

```
from collections import Counter
```

```
from sklearn.preprocessing import LabelEncoder  
from sklearn.preprocessing import MinMaxScaler  
from sklearn.feature_selection import SelectKBest  
from sklearn.feature_selection import mutual_info_classif  
from sklearn.model_selection import RandomizedSearchCV  
from sklearn.model_selection import GridSearchCV
```

```
import xgboost as xgb  
from xgboost import XGBClassifier
```

```
from sklearn.ensemble import RandomForestClassifier  
from sklearn.metrics import classification_report, confusion_matrix,  
accuracy_score
```

```
from sklearn.tree import DecisionTreeClassifier, export_graphviz  
from sklearn.metrics import classification_report, accuracy_score,  
confusion_matrix  
from sklearn import metrics
```

```
#from google.colab import drive  
#drive.mount('/content/drive')
```

```
# train = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/cse498r/human  
activities/train.csv') #zara's path  
# test = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/cse498r/human  
activities/test.csv') #zara's path
```

```
#train = pd.read_csv('/content/drive/MyDrive/Colab  
Notebooks/CSE498R/train.csv') #Baker's Path  
#test = pd.read_csv('/content/drive/MyDrive/Colab  
Notebooks/CSE498R/test.csv') #Baker's Path
```

```
#train = pd.read_csv('/content/drive/MyDrive/final dataset/train.csv')  
#test = pd.read_csv('/content/drive/MyDrive/final dataset/test.csv')  
train = pd.read_csv('E:/CSE498R/final dataset/test.csv')  
test = pd.read_csv('E:/CSE498R/final dataset/train.csv')
```

```
#see what is present in the data
```

```

train.head()

#see what is present in the data
train.tail()

train.info()
Train.shape
train.describe()
Train.dtypes

# how many records are available for each subjects
train.subject.value_counts()

#count how many unique value every feature have
train.nunique()

train['Activity'].unique()

print('Number of duplicates in train : ',sum(train.duplicated()))
print('Number of duplicates in test : ',sum(test.duplicated()))

print('Number of Missing values in train : ',train.isna().values.sum())
print('Number of Missing values in test : ',test.isna().values.sum())

train.isnull().sum()
train['Activity'].value_counts()
sns.countplot(x=train['Activity'],data=train,order
=train['Activity'].value_counts().index)
plt.title('Distribution of Activity(target) classes')
plt.xticks(rotation=50)
plt.show()

train.columns

#assigning label encoder
le = LabelEncoder()
#label encoding target class "Activity"
train[['Activity']] = train[['Activity']].apply(le.fit_transform)

print(train.shape)

```

```
train.head(10)
```

```
train.tail(20)
```

```
train.head()
```

```
#Counting how much parameter does each feature has
pd.DataFrame.from_dict(Counter([col.split('-')[0].split('(')[0]
                               for col in train.columns]),orient =
'index').rename(columns={0:'count'}).sort_values('count',ascending = False)
```

```
facetGrid = sns.FacetGrid(train,hue='Activity',height=5,aspect=3)
facetGrid.map(sns.distplot,'tBodyAccMag-mean()',hist = False).add_legend()
```

```
plt.annotate("Static Activities", xy = (-.98, 8), xytext = (-.8, 16),
arrowprops={'arrowstyle': '-', 'ls': 'dashed'})
plt.annotate("Static Activities", xy = (-.98, 13), xytext = (-.8, 16),
arrowprops={'arrowstyle': '-', 'ls': 'dashed'})
plt.annotate("Static Activities", xy = (-.98, 16), xytext = (-.8, 16),
arrowprops={'arrowstyle': '-', 'ls': 'dashed'})
```

```
plt.annotate("Dynamic Activities", xy=(-0.2,3.25), xytext=(0.1,
9),arrowprops={'arrowstyle': '-', 'ls': 'dashed'})
plt.annotate("Dynamic Activities", xy=(0.1,2.18), xytext=(0.1,
9),arrowprops={'arrowstyle': '-', 'ls': 'dashed'})
plt.annotate("Dynamic Activities", xy=(-0.01,2.15), xytext=(0.1,
9),arrowprops={'arrowstyle': '-', 'ls': 'dashed'})
sns.boxplot(x='Activity',y='tBodyAccMag-mean()',data=train, showfliers=False)
plt.ylabel("Body Accelerationmagnitude mean")
plt.title('Boxplot of tBodyAccmag-mean() column across various activities')
plt.axhline(y=-0.8,xmin=0.01,dashes=(3,3))
plt.show()
```

```
#Splitting the dataset
```

```
y = train['Activity']
X = train.drop(['Activity'], axis = 'columns')      #test data (removing target
column)
```

```
y.head(10)
```

```
X.head(10)
```

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2,  
random_state = 20, stratify = y)
```

```
print(X_train.shape)
```

```
print(X_test.shape)
```

```
train.columns
```

```
X_train.head()
```

```
scaler = MinMaxScaler().fit(X_train)
```

```
print(scaler)
```

```
scaler.data_min_
```

```
scaler.data_max_
```

```
X_train.describe()
```

```
scaler.feature_range
```

```
scaler.transform(X_train)
```

```
scaler = MinMaxScaler().fit(X_test)
```

```
print(scaler)
```

```
scaler.data_min_
```

```
# create a decision tree model
```

```
dt = DecisionTreeClassifier()
```

```
dt.fit(X_train, y_train) # train with training set; fit function is used to train the  
model
```

```
y_pred = dt.predict(X_test)
```

```
print(classification_report(y_test, y_pred))
```

```
print("Train Accuracy: ", accuracy_score(y_train, dt.predict(X_train)))
```

```
print("Test Accuracy: ", accuracy_score(y_test, y_pred))

# Initialize the XGBoost classifier
xgb_classifier = XGBClassifier(random_state=42)

# Fit the classifier on the training data
xgb_classifier.fit(X_train, y_train)

# Predict on the test data
y_pred_xgb = xgb_classifier.predict(X_test)

# Calculate and print the accuracy
accuracy_xgb = accuracy_score(y_test, y_pred_xgb)
print("XGBoost Model Accuracy: {:.2f}%".format(accuracy_xgb * 100))

# Generate and print the classification report
class_report_xgb = classification_report(y_test, y_pred_xgb)
print("Classification Report:\n", class_report_xgb)

# Generate and print the confusion matrix
confusion_mat_xgb = confusion_matrix(y_test, y_pred_xgb)
print("Confusion Matrix:")
print(confusion_mat_xgb)
```

-----**The End**-----